

Shangmin Chen

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

September 2022 – June 2026

EXPERIENCE

Software Engineer

BU Spark!

September 2024 – December 2024

Boston, MA

- Developed full-stack food waste platform using Django, JavaScript, and SQLite, connecting 500+ students and reducing campus waste by 10%
- Built RESTful API with sub-100ms query times and deployed via CI/CD pipelines on Heroku
- Implemented Auth0 authentication and Django security middleware, ensuring zero security incidents
- Accelerated release by 1 week using Agile methodology in team of 5, contributing to 7 iterations

IT Consultant

Boston University Engineering IT

January 2024 – July 2025

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Audio Transcription Platform | *Java Spring Boot, Python FastAPI, React, PostgreSQL*

September 2025

- Built microservices platform with Spring Boot API, Python FastAPI service, and React frontend, implementing async audio processing with OpenAI Whisper and real-time status polling
- Designed PostgreSQL database with 4 related tables, Flyway migrations, and optimized indexing, supporting multiple audio formats with comprehensive error handling
- Deployed containerized application using Docker Compose with inter-service communication, CORS configuration, and health monitoring endpoints

Fitness Tracking Mobile App | *React Native, TypeScript, Expo, Supabase*

August 2025

- Architected cross-platform mobile application with secure OAuth authentication, implementing token management and session persistence for seamless user experience across iOS and Android
- Built scalable backend integration with Supabase PostgreSQL, designing type-safe database services with Row Level Security policies and real-time data synchronization for multi-user workout tracking
- Engineered responsive UI with React Native Reanimated v3 for smooth gesture-driven interactions, implementing custom state management hooks and offline-first architecture with intelligent caching strategies

Crime Analytics & Forecasting Platform | *Python, Streamlit, Prophet, Scikit-learn, GeoPandas*

December 2024

- Built crime analytics platform processing 500K+ Boston crime records with Prophet time-series forecasting and DBSCAN clustering for hotspot identification across 12+ police districts
- Implemented advanced data pipeline with seasonal decomposition and confidence intervals, generating 2-month crime predictions with train/test validation
- Developed interactive dashboard using Streamlit and Plotly, creating 121 crime-type heat maps with Folium and real-time geospatial visualizations

Decentralized ML Model Marketplace | *Solidity, Circom, IPFS, Express.js, TensorFlow*

May 2024

- Built decentralized marketplace using Solidity smart contracts with ERC-20 tokens, implementing 6-state transaction management, automated candidate selection, and economic incentive mechanisms with stake slashing
- Integrated IPFS distributed storage with Ethereum blockchain using Thirdweb SDK, enabling secure model uploads with SHA256 hash verification and automated ML training workflows

- Implemented zero-knowledge proof system using Circom and Groth16 protocol for privacy-preserving accuracy verification, with MetaMask integration and real-time blockchain event monitoring

Self-Hosted Media & Application Server | *Docker, Linux, Cloudflare, Nginx, Automation*

Ongoing

- Architected containerized home server infrastructure using Docker Compose, orchestrating 10+ microservices including Sonarr, Radarr, Prowlarr, Jellyfin, Jellyseerr, and qBittorrent for automated media management and streaming with 99%+ uptime
- Implemented secure external access using Cloudflare Tunnels with zero-trust network architecture, eliminating port forwarding vulnerabilities while hosting personal website and services accessible via custom domain
- Configured automated workflow pipelines with inter-container networking and reverse proxy routing, integrating Slskd for distributed music acquisition and establishing centralized monitoring for resource optimization

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS, Solidity

Frameworks & Libraries: React, React Native, Node.js, Spring Boot, FastAPI, Django, Streamlit, Prophet, Scikit-learn

Technologies & Tools: Docker, Git, PostgreSQL, SQLite, IPFS, Auth0, Heroku, OpenAI API, GeoPandas, Plotly

Platforms & Systems: AWS, Ethereum, MetaMask, CrowdStrike, Microsoft Deployment Toolkit, NAS